Ch 10.3: Convolutional Neural Nets

Lecture 31 - CMSE 381

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Fri, Nov 22, 2024

Announcements

Last time:

- Multilayer NN
- pyTorch

This lecture:

- CNNs

Announcements:

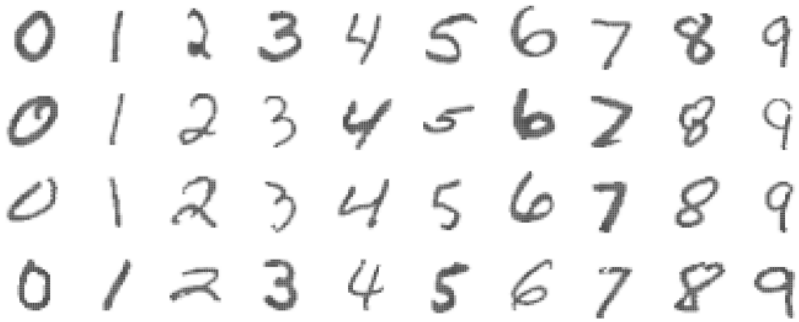
- Project due Friday
- HW #9 is posted
- Exam 3 is Dec 4th
- Project is due Dec 6th

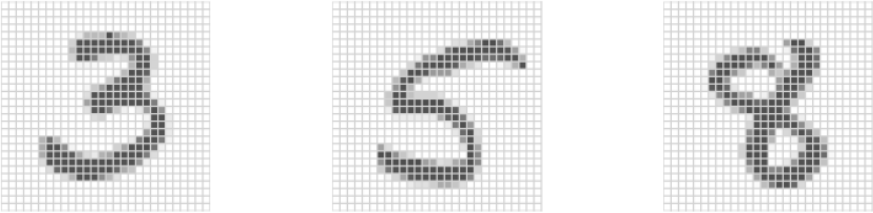
Lec #	Date			Reading	HW
21	Mon	10/28	Polynomial & Step Functions	7.1,7.2	
22	Wed	10/30	Step Functions; Basis functions; Start Splines	7.2 - 7.4	
23	Fri	11/1	Regression Splines	7.4	HW #6 Due Sun 11/3
24	Mon	11/4	Decision Trees	8.1	
25	Wed	11/6	Class Cancelled (Dr Munch out of town)		
26	Fri	11/8	Random Forests	8.2.1, 8.2.2	HW #7 Due Sun 11/10
27	Mon	11/11	Maximal Margin Classifier	9.1	
28	Wed	11/13	SVC	9.2	
29	Fri	11/15	SVM	9.3, 9.4	HW #8 Due Sun 11/17
30	Mon	11/18	Single layer NN	10.1	
31	Wed	11/20	Multi Layer NN	10.2	
32	Fri	11/22	CNN	10.3	HW #11 Due Sun 11/24
33	Mon	11/25	TBD: Unsupervised learning/clustering	12.1, 12.4?	
	Wed	11/27	Virtual: Project office hours		
	Fri	11/29	No class - Thanksgiving		
	Mon	12/2	Review		
	Wed	12/4	**Midterm #3**		
	Fri	12/6	No class - EGR Design Day		Project due

Section 1

Last time: Neural Nets

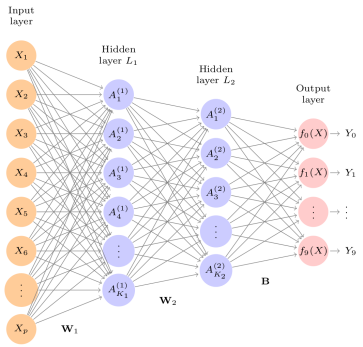
MNIST





- Goal: Build a model to classify images into their correct digit class
- Each image has $p = 28 \cdot 28 = 784$ pixels
- Each pixel is grayscale value in [0,255]
- Data converted into column order
- Output represented by one-hot vector $Y = (Y_0, Y_1, \cdots, Y_9)$
- 60K training images, 10K test images

Neural network architecture for MNIST

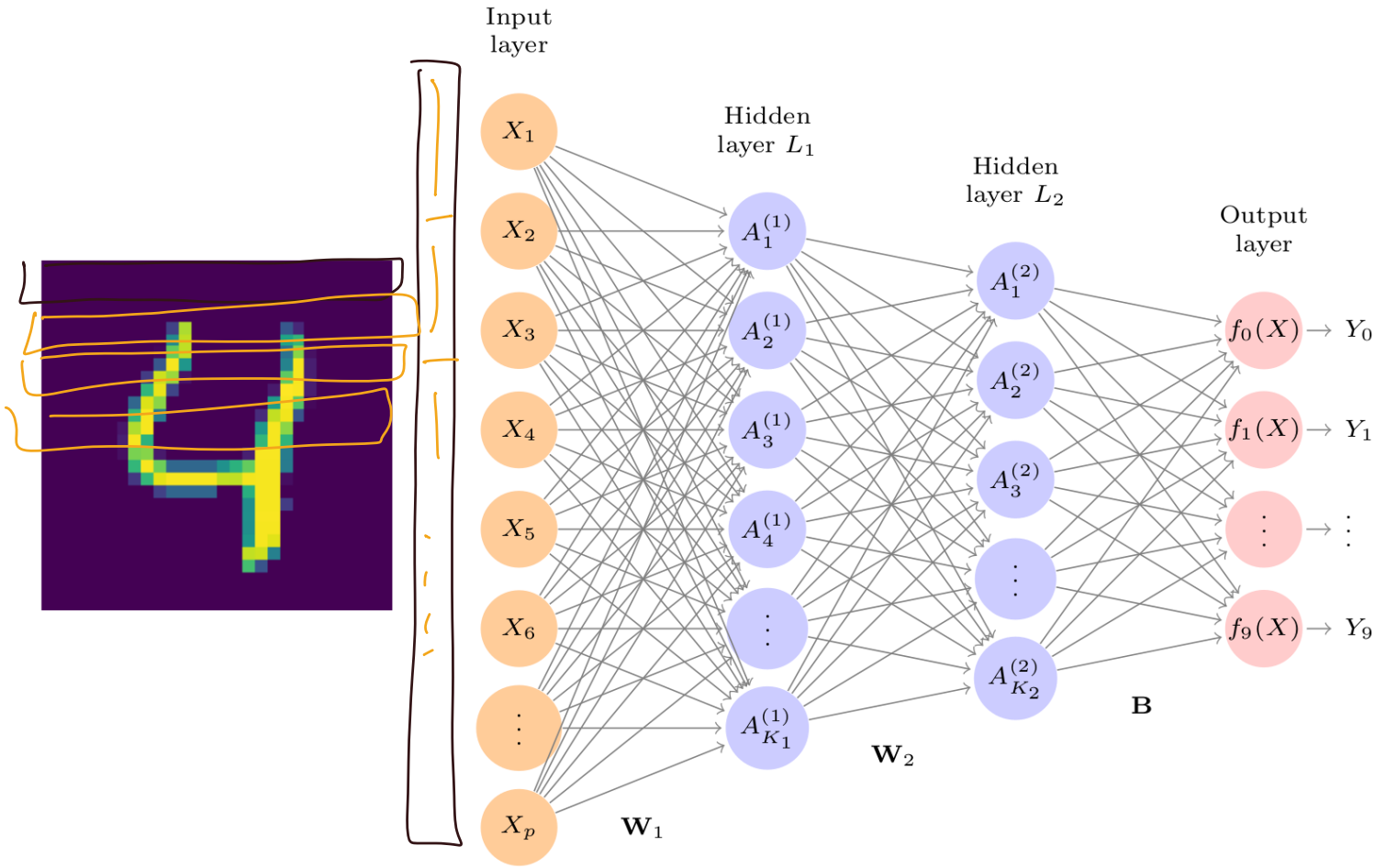


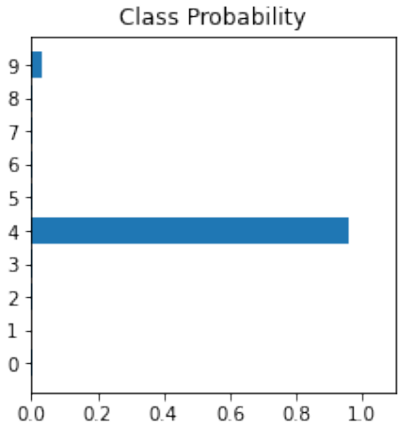
$$
\begin{aligned}
A_k^{(1)} &= h_k^{(1)}(X) \\
&= g(w_{k0}^{(1)} + \textstyle\sum_{j=1}^{p} w_{kj}^{(1)} X_j) \\
A_\ell^{(2)} &= h_\ell^{(2)}(X) \\
&= g(w_{\ell 0}^{(2)} + \textstyle\sum_{k=1}^{K_1} w_{\ell k}^{(2)} A_k^{(1)}) \\
Z_m &= \beta_{m0} + \textstyle\sum_{\ell=1}^{K_2} \beta_{m\ell} h_\ell^{(2)}(X) \\
&= \beta_{m0} + \textstyle\sum_{\ell=1}^{K_2} \beta_{m\ell} A_\ell^{(2)},
\end{aligned}
$$

$$
f_m(X) = \Pr(Y = m|X) = \frac{e^{Z_m}}{\sum_{\ell=0}^{9} e^{Z_\ell}},
$$

- Two hidden layers.
- Softmax for classification output
- We used L_1 has 128 units; L_2 has 64
- 10 output variables due to class labeling
- Result is we are training approx 110K weights

MNIST learning





Section 2

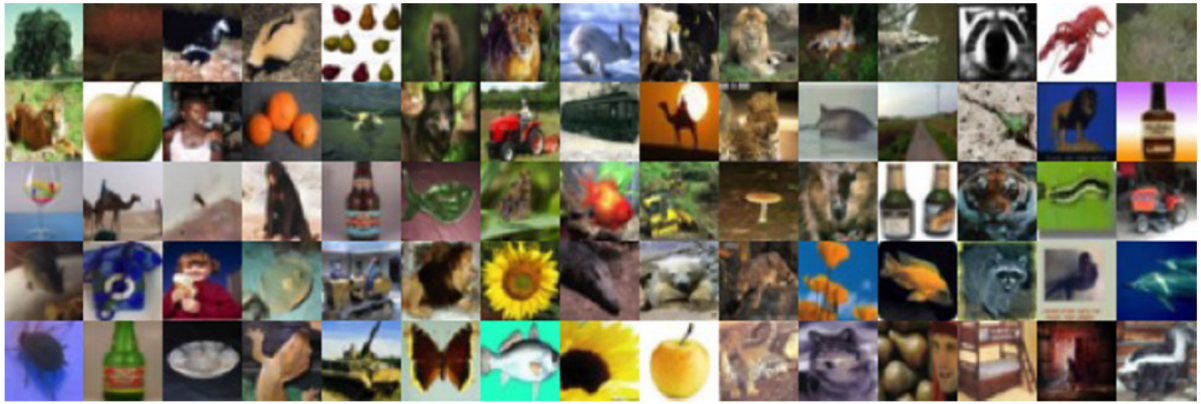
Convolutional Neural Network

Last time: Flattening the image

$$\begin{pmatrix} 1 & 1 & 0 \\ 4 & 2 & 1 \\ 0 & 2 & 1 \end{pmatrix} \longrightarrow \begin{pmatrix} 1 \\ 1 \\ 0 \\ 4 \\ 2 \\ 1 \\ 0 \\ 2 \\ 1 \end{pmatrix}$$

- Ignored that the input was an image.
- Lost structure might have been helpful
- Goal CNN: Keep the matrix structure

Example data set: CIFAR100 Data



- 60,000 images: 50K training, 10K test
- Labels with 20 super classes (e.g. aquatic mammals)
- 5 classes per super class (beaver, dolphin, otter, seal, whale)
- Images are 32x32

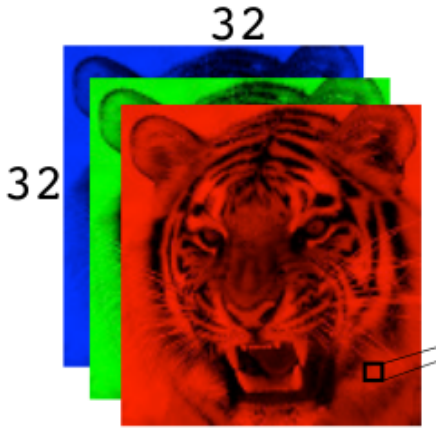
Image channel data

3 #'s per pixel → (R, G, B) values

Tensor: multi dim matrix

2 spatial axes 32x32 pixels

channel axis → rep's the three colors



CNNs

label

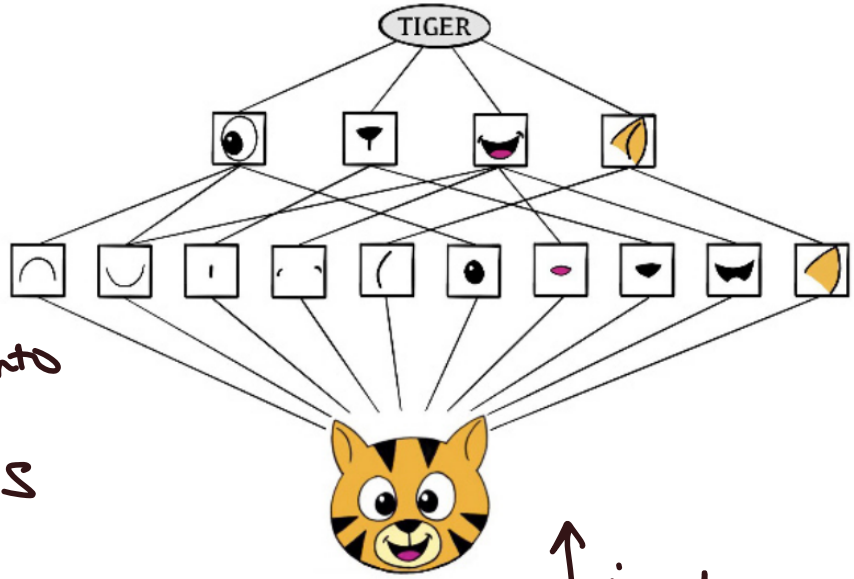
↑

high level features

↑

↑ combined into

Low level features



↑ input

Two kinds of layers

- Convolution layers
- Pooling layers

Convolution layer

Convolution Filter

Original Image:

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \\ j & k & l \end{bmatrix}$$

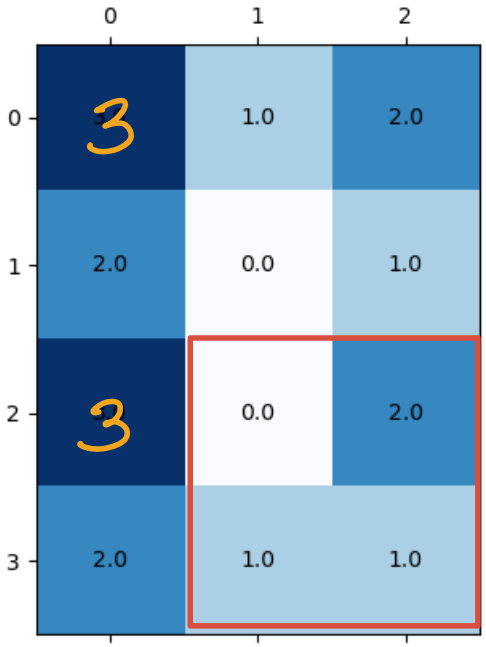
Convolution filter:

$$\boxed{\begin{bmatrix} \alpha & \beta \\ \gamma & \delta \end{bmatrix}}$$

Convolved Image

$$\begin{bmatrix} a\alpha + b\beta + d\gamma + e\delta & b\alpha + c\beta + e\gamma + f\delta \\ d\alpha + e\beta + g\gamma + h\delta & e\alpha + f\beta + h\gamma + i\delta \\ g\alpha + h\beta + j\gamma + k\delta & h\alpha + i\beta + k\gamma + l\delta \end{bmatrix}$$

Convolution Filter Example



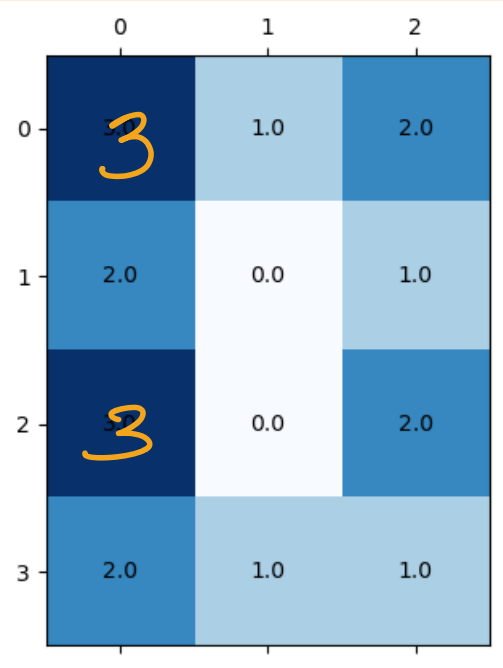
Filter:

$$\begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 \\ 0 & 3 \\ 1 & 3 \end{bmatrix}$$

Same example, different filter

What is the convolved image?

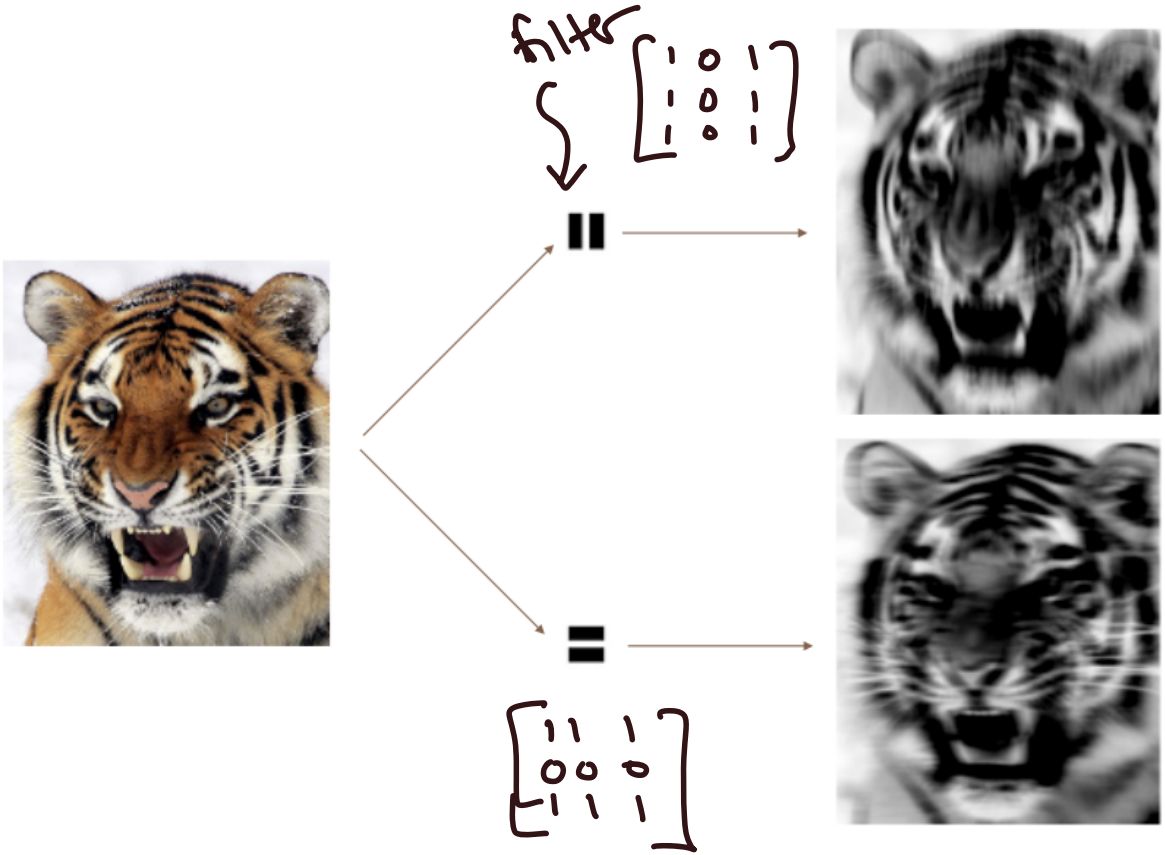


Filter:

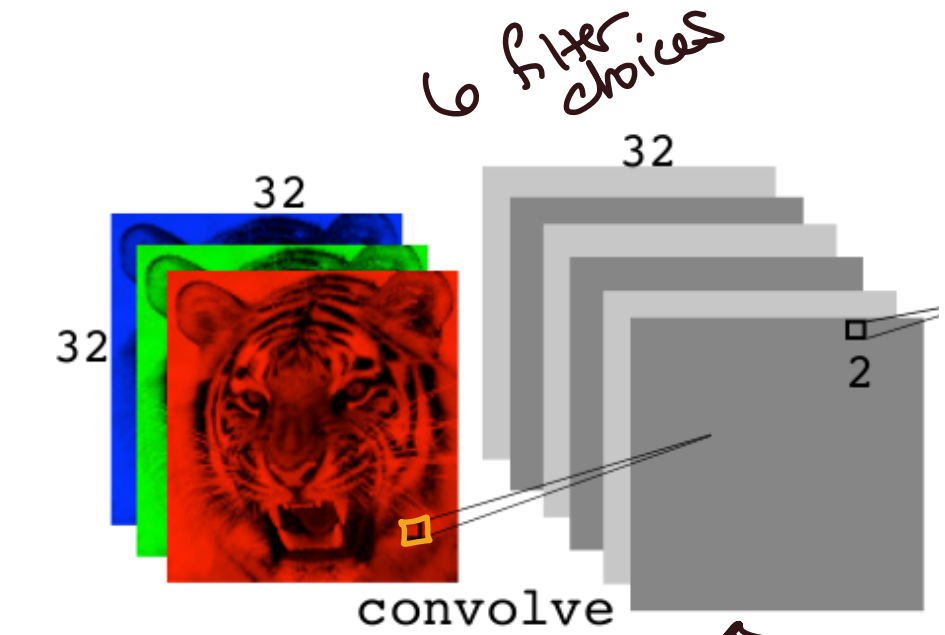
$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 3 \\ 2 & 1 \\ 3 & 2 \end{bmatrix}$$

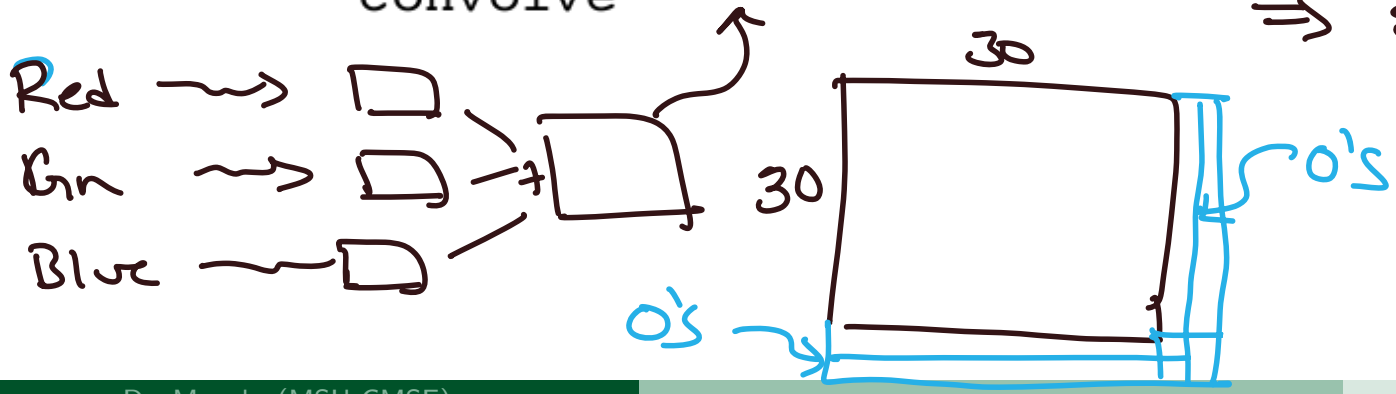
Convolution filter: Bigger example

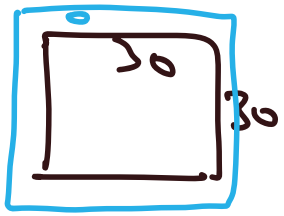


Convolution layer



- Convolution filter have 3 channels, one per color, each of size 3x3
- Results are summed to single 2-dim output
- Padding of convolved image ⇒ same size pixel output





More notes on convolution

- Idea of filters from image processing
  - ↳ Novelty in CNN is that the weights are learned
- Common to apply ReLU or other activation fcn in the layer.
  - ↳ Sometimes called separate layer "detector layer"

Pooling layers

Condense matrix to smaller matrix

Each 2x2 non-overlapping block

2x2

Max pool $\begin{bmatrix} 1 & 2 & 5 & 3 \\ 3 & 0 & 1 & 2 \\ 2 & 1 & 3 & 4 \\ 1 & 1 & 2 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 3 & 5 \\ 2 & 4 \end{bmatrix}$

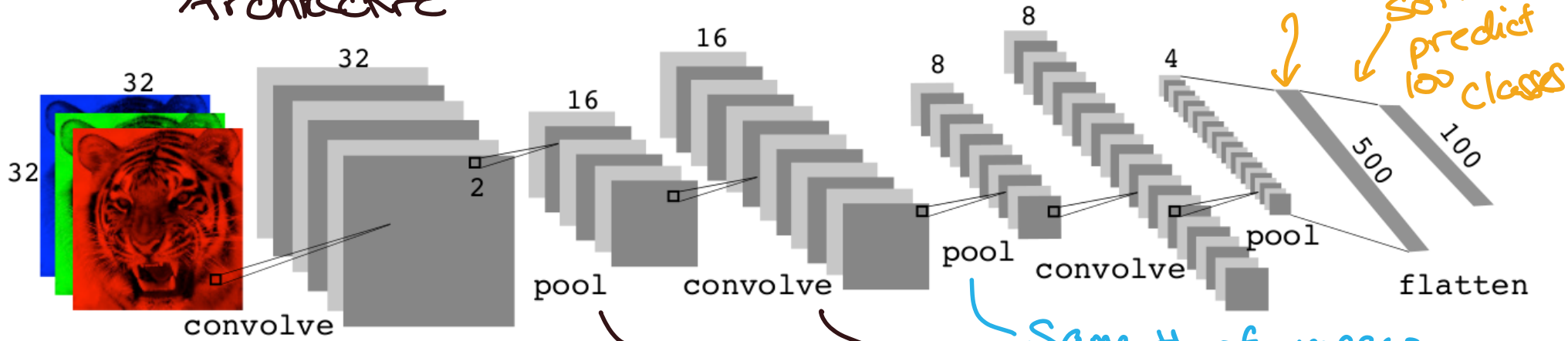
makes a $n \times n$ matrix into $\frac{n}{2}$ by $\frac{n}{2}$ matrix

Provides location invariance:

A large value in the region is maintained

Putting it together to make a CNN

Architecture



Soft max predict 100 classes

Same # of images smaller res.

max pool

B/c smaller images, chose to use more filters

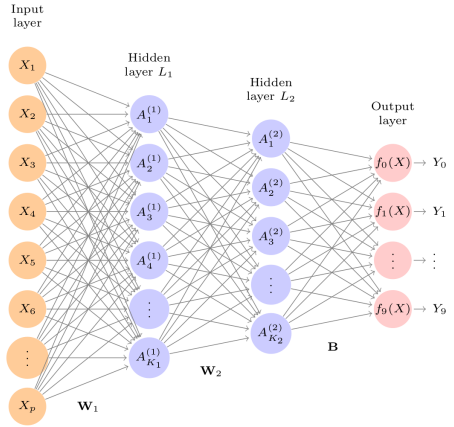
- Start w/ 3 channels
- Each filter in 1st layer → 1 convolved image

https://poloclub.github.io/cnn-explainer/

Coding

TL;DR

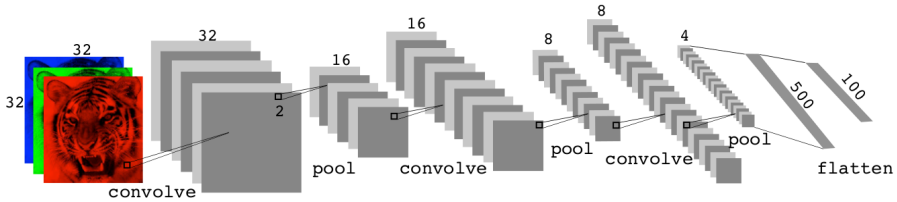
Feed Forward Neural Net



$$A_k = h_k(X) = g(w_{k0} + \sum_{j=1}^{p} w_{kj} X_j),$$

- Combines input data using learned weights
- Linear combo of those to get output
- Sometimes softmax to get probability of classification

CNN



- Specialized NN
- Gets next layer via
  - Convolution layer
  - Pooling Layer
  - Fully connected layer

Next time

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